

COLUMBUS GLENN, OH, USA

WMO: 724280

Lat: 39.991N

Lon: 82.877W

Elev: 249

StdP: 98.37

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14821

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.3	-12.3	-20.4	0.6	-14.1	-17.7	0.8	-11.5	12.0	0.3	11.0	0.1	3.3	260	0.487

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	32.8	23.0	31.6	22.5	30.3	21.9	24.7	30.3	24.0	29.1	23.2	28.0	4.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	18.4	27.1	22.4	17.6	26.6	21.7	16.8	26.0	75.9	30.4	73.0	29.3	70.1	27.9	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	8.9	7.9	-18.3	34.6	4.0	1.5	-21.2	35.7	-23.6	36.5	-25.8	37.4	-28.8	38.5		
(5)			-18.8	26.1	3.9	0.9	-21.6	26.8	-23.9	27.3	-26.1	27.8	-28.9	28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.2	-1.5	0.4	5.4	12.0	17.5	22.4	24.1	23.5	19.8	13.2	6.6	1.5
	DBStd	10.29	6.88	6.30	6.11	5.15	4.61	3.39	2.74	2.68	3.87	4.85	5.20	5.59
	HDD10.0	1260	360	275	169	37	2	0	0	0	0	21	127	268
	HDD18.3	2867	613	503	401	196	73	7	0	1	28	172	352	521
	CDD10.0	2046	5	5	28	98	236	371	437	418	296	121	26	6
	CDD18.3	610	0	0	1	7	48	128	178	161	73	13	0	0
	CDH23.3	5261	0	1	11	73	427	1104	1602	1339	601	102	1	0
	CDH26.7	1744	0	0	1	8	114	383	584	445	187	22	0	0
Wind	WSAvg	3.5	4.2	4.1	4.1	4.1	3.4	3.1	2.8	2.6	2.7	3.2	3.6	3.8
Precipitation	PrecAvg	996	65	57	84	89	100	103	113	95	76	62	75	75
	PrecMax	1433	228	133	258	182	250	264	314	291	185	170	271	177
	PrecMin	623	17	7	18	17	24	16	29	19	10	3	11	22
	PrecStd	169	36	28	44	39	45	50	53	51	44	34	42	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.9	19.7	24.7	27.9	31.3	33.8	34.9	33.8	33.2	29.3	21.9	17.9
		MCWB	14.3	14.6	15.9	18.1	21.3	23.0	24.8	23.3	22.1	20.5	15.7	15.0
	2%	DB	14.6	15.5	21.3	25.5	29.5	32.0	32.8	32.1	30.9	26.1	19.5	14.8
		MCWB	12.6	11.8	14.0	16.3	19.9	22.4	23.7	22.9	21.4	18.6	14.5	12.3
	5%	DB	11.8	12.7	18.1	23.2	27.9	30.5	31.4	30.7	28.8	23.7	17.2	12.4
		MCWB	9.9	9.5	12.4	15.2	19.5	21.6	22.9	22.4	20.7	17.1	13.1	10.4
	10%	DB	8.4	9.6	15.3	20.9	25.9	29.0	30.0	29.2	26.8	21.4	15.1	9.9
		MCWB	6.1	6.5	10.8	13.8	18.7	21.0	22.1	21.7	19.6	16.5	11.4	7.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.9	15.5	17.1	19.7	22.6	24.7	26.2	25.6	24.0	21.8	17.5	15.5
		MCDB	17.0	19.2	23.4	25.8	29.0	31.0	32.0	31.0	29.6	27.2	20.3	17.5
	2%	WB	13.0	12.6	15.1	17.9	21.5	23.7	25.0	24.5	22.9	20.2	15.5	13.0
		MCDB	14.6	14.7	19.4	23.0	27.2	29.6	30.9	29.5	28.0	24.0	18.4	14.2
	5%	WB	10.2	9.8	13.5	16.5	20.5	22.9	24.1	23.8	22.0	18.6	13.8	10.6
		MCDB	11.6	12.6	16.9	20.8	25.7	28.4	29.2	28.4	26.7	22.5	16.5	12.2
	10%	WB	6.5	6.7	11.4	15.2	19.6	22.1	23.4	22.9	21.1	16.9	11.8	7.7
		MCDB	8.3	9.4	15.4	19.6	24.4	27.2	27.9	27.0	25.1	20.7	14.6	9.6
Mean Daily Temperature Range	5% DB	MDBR	7.7	8.4	9.7	11.2	10.7	10.2	10.0	10.0	10.7	10.3	8.8	7.1
		MCDBR	9.5	12.6	13.7	13.8	12.8	12.0	11.3	11.4	12.7	13.1	12.1	10.1
		MCWBR	8.2	9.9	8.3	7.7	5.6	4.8	4.1	4.3	5.0	6.6	8.3	8.7
	5% WB	MCDBR	9.2	11.6	11.6	11.5	10.7	10.3	10.0	9.6	10.3	10.8	10.2	9.6
		MCWBR	8.4	10.0	8.1	7.4	5.4	4.7	4.3	4.3	4.6	6.1	8.1	9.0
Clear-Sky Solar Irradiance	taub	0.310	0.330	0.360	0.397	0.438	0.456	0.465	0.456	0.397	0.365	0.328	0.306	
	taud	2.422	2.345	2.336	2.254	2.175	2.168	2.154	2.162	2.305	2.387	2.452	2.456	
	Ebn at Noon	853	883	892	877	845	827	815	812	841	832	824	829	
	Edh at Noon	82	101	114	133	147	148	149	144	117	97	79	74	
All-Sky Solar Radiation	RadAvg	1.63	2.48	3.48	4.71	5.48	6.00	5.86	5.42	4.42	3.05	2.06	1.39	
	RadStd	0.16	0.32	0.32	0.41	0.46	0.42	0.34	0.28	0.40	0.27	0.27	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+115	+67
(47) Regional (42 neighbors)		+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+121	+67

Nomenclature: See separate page